

WOENS DRECHT, Netherlands

WMO: 063400

Lat: 51.4478N

Lon: 4.3419E

Elev: 63

StdP: 14.66

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	23.3	13.6	11.0	22.1	18.0	13.6	25.0	28.7	49.3	24.4	48.1	5.8	70	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	84.7	67.7	80.4	66.3	76.9	64.6	69.8	80.4	68.0	77.3	66.3	74.5	7.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.3	97.1	73.3	64.6	91.4	71.5	63.2	86.8	70.1	33.8	80.4	32.3	77.5	31.0	74.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	18.9	16.7	DB	12.7	91.8	5.2	3.0	9.0	93.9	5.9	95.6	3.0	97.3	-0.8	99.5	(4)
(5)			WB	12.2	73.5	5.2	1.3	8.5	74.4	5.4	75.2	2.5	75.9	-1.2	76.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.0	38.2	39.2	43.4	49.4	55.5	60.7	64.4	64.1	59.0	52.5	44.8	39.6		
	DBStd	11.37	7.97	7.23	6.31	6.62	6.32	5.63	5.35	4.96	5.00	6.23	6.45	7.56		
	HDD50	1553	369	304	216	90	20	0	0	0	1	46	178	329		
	HDD65	5271	831	721	668	469	301	153	77	78	190	388	606	789		
	CDD50	1907	2	3	13	72	190	322	446	437	271	124	21	6		
	CDD65	150	0	0	0	1	6	25	57	50	10	1	0	0		
	CDH74	1466	0	0	0	24	102	284	522	450	80	4	0	0		
	CDH80	396	0	0	0	2	14	76	156	135	13	0	0	0		
(14) Wind	WSAvg	7.8	8.9	8.8	8.6	7.7	7.9	7.5	7.3	6.8	6.7	7.3	7.4	8.6		
(15) Precipitation	PrecAvg	33.3	2.7	2.4	1.9	1.7	2.4	2.9	3.4	3.3	3.0	2.8	3.3	3.4		
	PrecMax	44.1	5.6	5.0	4.3	3.8	4.2	4.8	7.0	8.6	8.9	6.7	5.2	7.0		
	PrecMin	24.1	0.2	0.7	0.3	0.0	0.7	1.1	0.8	0.6	0.5	0.9	0.4	1.0		
	PrecStd	4.4	1.3	1.0	1.1	0.9	1.0	1.2	1.5	1.9	2.0	1.2	1.2	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	58.4	65.9	77.1	82.2	87.5	90.3	88.9	81.4	73.1	62.6	56.7		
		MCWB	52.0	49.2	54.0	59.0	65.8	68.1	69.9	69.5	66.6	63.3	57.6	53.2		
	2%	DB	53.0	54.0	60.8	71.3	77.0	81.3	84.4	84.0	75.6	67.5	58.8	54.5		
		MCWB	50.2	49.4	52.2	56.7	61.3	66.2	67.6	67.9	64.9	60.9	55.1	51.8		
	5%	DB	50.7	51.7	56.6	65.9	72.7	76.7	79.6	79.0	71.6	64.1	56.3	52.3		
		MCWB	48.1	48.2	50.2	54.6	60.0	64.3	66.5	66.5	62.6	59.1	53.1	49.9		
	10%	DB	48.5	49.5	53.6	61.2	68.2	72.4	75.5	74.4	68.1	61.6	54.0	49.9		
		MCWB	46.3	46.8	48.8	52.5	58.5	62.6	65.0	64.7	60.8	57.6	51.2	47.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	52.1	55.8	61.1	67.5	71.0	71.9	72.2	68.6	64.2	58.2	54.3		
		MCDB	54.6	55.4	62.2	73.5	78.0	82.2	83.9	83.5	77.7	70.6	60.9	55.7		
	2%	WB	50.5	50.3	53.2	57.7	63.9	67.9	69.8	69.6	66.0	61.7	55.5	52.2		
		MCDB	52.6	52.8	58.6	67.4	72.5	77.3	80.6	80.0	73.4	66.2	58.1	54.1		
	5%	WB	48.4	48.7	51.2	55.5	61.3	65.7	67.9	67.5	63.9	59.9	53.6	50.2		
		MCDB	50.4	51.2	55.5	64.1	69.7	74.9	77.4	76.2	69.5	63.5	56.1	52.2		
	10%	WB	46.5	46.8	49.4	53.5	59.1	63.5	66.0	65.8	62.2	58.2	51.4	47.8		
		MCDB	48.4	49.5	53.1	60.0	66.9	70.5	73.5	73.0	66.9	61.2	53.6	49.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	11.2	14.5	18.1	17.9	17.8	17.0	17.5	16.3	13.6	10.5	8.7		
		MCDBR	9.3	13.3	20.4	25.9	26.1	25.6	24.9	25.2	20.9	15.7	11.0	8.2		
		MCWBR	8.3	9.8	13.2	13.4	12.8	12.8	11.2	11.6	11.3	10.0	8.4	7.3		
		MCDBR	9.1	10.9	17.0	22.3	21.5	22.2	21.3	21.7	18.1	14.7	10.7	8.3		
	5% WB	MCWBR	8.5	9.0	12.0	12.4	12.0	12.2	10.8	11.1	10.6	10.1	8.8	7.7		
(40) Clear-Sky Solar Irradiance	taub		0.335	0.357	0.391	0.422	0.431	0.427	0.433	0.423	0.398	0.372	0.349	0.333		
	taud		2.354	2.330	2.239	2.186	2.208	2.242	2.230	2.275	2.336	2.389	2.369	2.331		
	Ebn at Noon		211	239	252	258	261	264	259	255	248	231	204	190		
	Edh at Noon		20	27	35	42	43	42	42	38	32	25	20	18		
(44) All-Sky Solar Radiation	RadAvg		238	455	828	1335	1597	1727	1621	1384	1021	596	278	182		
	RadStd		27	87	115	139	129	138	189	121	89	48	28	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page